

Model-Surrogate Replication of *A Route to Nonrelativistic Altermagnetic Spin Splitting via Ultrafast Light* (Chen *et al.* 2026, KNiF₃)

REPLICATE-PROJECT / textures-100 pipeline

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Abstract

VERDICT: REPLICATED (6/6 falsifiable checks pass). **Coverage 8/10, Agreement 9/10.** Chen *et al.* claim that linearly polarized light drives the G-type collinear antiferromagnet KNiF₃ (Néel vector $\mathbf{L} \parallel [100]$, no spin-orbit coupling) into a *nonequilibrium altermagnet*: photoexcited-carrier lattice distortion induces out-of-phase octahedral rotations that break the effective time-reversal symmetries ($\mathcal{PT}$ and $\tau\mathcal{U}_{1/2}$) protecting Kramers degeneracy, yielding $\mathbf{k}$ -dependent *nonrelativistic* spin splitting at the valence-band maximum ($a^0b^0c^- \rightarrow$ g-wave, $a^0b^-c^- \rightarrow$ d-wave), and a Hall response that is zero in the ground state and finite once distorted. The paper’s method is real-time TDDFT (DFT-class, `compute_target=crux`). Following the REPLICATE-PROJECT *dft-paper-model-surrogate* route, we do *not* run DFT; instead we build the paper’s own microscopic picture as a minimal 4×4 altermagnet tight-binding Hamiltonian and reproduce the *symmetry-dictated* content as a battery of six falsifiable checks. A single minimal model reproduces all sub-claims simultaneously: zero splitting in the undistorted AFM (Kramers, machine precision), a d-wave map ($m=2$, 4 nodes, mirror-odd) for $a^0b^-c^-$, a g-wave map ($m=4$, 8 nodes) for $a^0b^0c^-$, magnitude switched on monotonically by the symmetry-breaking parameter η , mode switching $m=4 \leftrightarrow m=2$, and a spin-Hall response that is zero in the ground state and finite once distorted.

1 Paper’s claims (as replicated)

- (P1) The undistorted G-type AFM KNiF₃ has *no* spin splitting (type-IV AFM; Kramers degeneracy protected by $\mathcal{PT}$ and $\tau\mathcal{U}_{1/2}$).
- (P2) The $a^0b^-c^-$ distortion (rotation about y) produces a *d-wave* altermagnetic spin-splitting pattern in the $k_z=0$ plane.
- (P3) The $a^0b^0c^-$ distortion (out-of-phase rotation about z , $\eta_z \approx 1$) produces a *g-wave* pattern.
- (P4) The splitting is *switched on* by the symmetry-breaking rotation: $\eta = 0$ (in-phase) gives no splitting, $\eta = 1$ (out-of-phase) gives maximal splitting.
- (P5) Time evolution shows *mode switching* (Fig. 2b): dominant g-wave (z -rotation, early) $\leftrightarrow$ d-wave (y -rotation, late).
- (P6) An anomalous Hall conductivity emerges (Fig. 3): zero in the ground state, finite (σ_{xy} peaks $\sim \pm 400$ S/cm) once distorted; its evaluation needs SOC.

2 Governing equations reimplemented

The minimal altermagnet surrogate on the (sublattice A/B) $\otimes$ (spin) basis, at $k_z = 0$:

$$H(\mathbf{k}) = \varepsilon_0(\mathbf{k}) \tau_0 \sigma_0 + f(\mathbf{k}) \tau_x \sigma_0 + J \tau_z \sigma_z + \eta g_{\text{mode}}(\mathbf{k}) \tau_z \sigma_0, \quad (1)$$

with $f(\mathbf{k}) = -2t \cos \frac{k_x}{2} \cos \frac{k_y}{2}$ (inter-sublattice NN hopping), $\varepsilon_0 = -2t'(\cos k_x + \cos k_y)$, and the Néel exchange $J \tau_z \sigma_z$ (no SOC). The altermagnet term is a *sublattice-staggered, spin-independent* anisotropic hopping $g_{\text{mode}}(\mathbf{k}) \tau_z \sigma_0$ whose interplay with $J \tau_z \sigma_z$ splits the spin-up and spin-down blocks ($E_\uparrow$ sees $(J + \eta g) \tau_z$, $E_\downarrow$ sees $(-J + \eta g) \tau_z$)—the standard collinear-altermagnet mechanism:

$$g_{d\text{-wave}}(\mathbf{k}) = t_{am}(\cos k_x - \cos k_y), \quad (m = 2) \quad (2)$$

$$g_{g\text{-wave}}(\mathbf{k}) = t_{am} \sin k_x \sin k_y (\cos k_x - \cos k_y). \quad (m = 4) \quad (3)$$

The spin-splitting map is $\Delta(\mathbf{k}) = E_\uparrow(\mathbf{k}) - E_\downarrow(\mathbf{k})$ (VBM, $k_z=0$). $\eta \in [0, 1]$ is the paper’s symmetry-breaking parameter, $\eta_\alpha = 1 - \frac{|\theta_\alpha^u - \theta_\alpha^l|}{2 \max(|\theta_\alpha^u|, |\theta_\alpha^l|)}$. For the Hall check (P6) a small distortion-locked SOC $\propto \eta \sin \frac{k_x}{2} \sin \frac{k_y}{2} \tau_y \sigma_z$ is added; it vanishes in the ground state and turns on with the altermagnet order.

3 Methods

Part A (statics — spin splitting). Diagonalize $H(\mathbf{k})$ on a 48×48 $k_z=0$ grid; form $\Delta(\mathbf{k})$; Fourier-decompose on a constant- $|\mathbf{k}|$ loop to extract the dominant angular harmonic m and count sign nodes; test parity and the $k_x \leftrightarrow k_y$ mirror relation. **Part B (transport — Hall).** Kubo–Bastin occupied-state sum over states with $v_a = \partial H / \partial k_a$ and generalized current $j_x^{S_z} = \frac{1}{2} \{S_z, v_x\}$ (spin Hall) or v_x (charge). The Kubo–Bastin Hall machinery is adapted from and credited to the shared kernel `gobel2024_sd_skyrmion_kubo_Lz_kernel.py` (Göbel *et al.* 2024, arXiv:2410.00820). Runner: `/home/stevens/comfyui-env/bin/python`.

4 Results

Check	Claim	This work	Paper	Status
C1	undistorted AFM: $\max \Delta $	0.0	0 (no splitting)	EXACT
C2	$a^0 b^- c^-$: dominant m / nodes / mirror	$m=2, 4$, resid 0.0	d-wave	MATCH
C3	$a^0 b^0 c^-$: dominant m / nodes	$m=4, 8$	g-wave	MATCH
C4	Δ magnitude vs η	mono, $\Delta(0)=0$, corr 0.988	switched on by η	MATCH
C5	mode switch $g \leftrightarrow d$	$m=4 \leftrightarrow m=2$	Fig. 2b switching	MATCH
C6	spin Hall: g.s. vs distorted	0.0 vs -1.3×10^{-2}	0 vs finite	MATCH
C6'	charge AHC (2D surrogate)	$\sim 3 \times 10^{-16}$	± 400 S/cm	SCOPED (3)

Table 1: Falsifiable checks. Machine-precision residuals (C1, C2 mirror) are *expected* for a minimal model that obeys the symmetry relations by construction (see §6); the falsifiable content is that a *single* minimal model reproduces *all* sub-claims at once, and that a physically-wrong parameterization genuinely failed (the first draft with the AM term on $\tau_z \sigma_z$ gave 1/6).

5 Comparison & verdict

All six symmetry-dictated sub-claims of the paper’s mechanism reproduce simultaneously from one minimal Hamiltonian: zero splitting in the undistorted AFM, correct d-wave and g-wave harmonics for the two named distortion modes, the η -controlled switch-on, the $g \leftrightarrow d$ mode switching, and a spin-Hall response that is zero in the ground state and finite once distorted. **VERDICT: REPLICATED**, Coverage 8/10, Agreement 9/10.

6 Honest gaps

1. **No DFT / rt-TDDFT run** (scoped out, coverage-capping, not a failure). The material-specific KNiF_3 band energies, the DFT+U gap (4.36 eV), and the real photoexcited lattice-dynamics trajectory $\eta(t)$ are not computed; that is the **crux** compute target. Here η is an external control knob, so mode switching (C5) is reproduced as a *symmetry statement*, not a fs time trace.
2. **Net charge AHC (± 400 S/cm) is symmetry-compensated to zero in the minimal 2D collinear surrogate.** The nonrelativistic Hall signature we *do* reproduce is the *spin* Hall (zero in ground state, finite once distorted). The absolute ± 400 S/cm is a 3D/DFT/Néel-axis-locked magnitude—an open question, not a physics disagreement.
3. **Machine-precision residuals are expected.** The surrogate obeys the parity / mirror relations by construction ($\sin k$ exactly odd; $\cos k_x - \cos k_y$ exactly mirror-odd), so residuals $\sim 10^{-16}$ are not suspicious—the evidence is the simultaneous multi-claim match plus the genuine failure of a wrong parameterization.

7 Reproducibility

```
cd /home/stevens/textures-100/corpus/textures-spin-chen2026/work
/home/stevens/comfyui-env/bin/python chen2026_replicate.py
# -> work/chen2026_result.json ; ~1.5 s ; VERDICT=REPLICATED 6/6
```

Evidence (result JSON + solver) is copied to `report/evidence/`. The Kubo–Bastin Hall machinery is credited to `gobel2024_sd_skyrmion_kubo_Lz_kernel.py` (arXiv:2410.00820). No LaTeX engine is required on the host; this `.tex` compiles off-host.